AORTHAR ACADEMY

Product, Engineering & QA Documentation

Comprehensive documentation pack for the dual-platform education product

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1. Full Product Overview

Project Summary

Stack: Next.js 16 App Router · TypeScript · Tailwind CSS v4 · shadcn/ui · Supabase · Paystack · Gemini AI · Resend

Deployment: Vercel (aorthar.com, courses.aorthar.com)

Last Updated: April 3, 2026

What This Product Is

Aorthar Academy is a **dual-platform education product**:

Platform 1: University (aorthar.com)

A structured 4-year academic program modeled after a university curriculum. Students enroll in a department, progress through year/semester courses, take quizzes/exams, earn grades, track GPA, and submit a capstone project. Freemium model — Years 100–300 are free, Year 400 + capstone require a premium subscription.

Platform 2: Standalone Courses (courses.aorthar.com)

A pay-per-course platform for self-paced learning. Individual courses with YouTube/Drive-powered lessons, lifetime access, one-time purchase via Paystack. Targets professionals who want specific skills without committing to the full university program.

User Types

PERSONA	DESCRIPTION
Student (University)	Enrolled in a department, progressing through the 4-year curriculum. Free tier + optional premium upgrade.
Student (Standalone)	Purchased one or more individual courses on courses.aorthar.com. No department enrollment required.
Contributor	A student who has had 3+ suggestions approved. Auto-promoted. Can create curriculum proposals.
Admin (Super)	Full access to all admin features, user management, pricing, audit logs, ops hub.
Admin (Content)	Can manage courses, lessons, questions, curriculum, suggestions, capstone reviews.
Admin (Finance)	Can view payments, transactions, audit logs, pricing configuration.
Visitor	Unauthenticated user browsing landing pages, pricing, course catalogs.

System Architecture

```
┌──────────────────────────────────────────────────────────┐
│                    Vercel (CDN + SSR)                    │
│  ┌──────────────────┐  ┌──────────────────┐             │
│  │  aorthar.com     │  │ courses.aorthar.com│           │
│  │  (University)    │  │ (Standalone)      │            │
│  │  Next.js 16      │  │ Next.js 16        │            │
│  └──────────┬───────┘  └──────────┬────────┘            │
│             └──────────┬──────────┘                      │
│                        ┌─────▼─────┐                     │
│                        │ Next.js App │                   │
│                        │ Router      │                   │
│                        └────────────┘                    │
└──────────────────────┬──────────────────────────────────┘
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              ┌────────┼────────┐
              │        │        │
      ┌───────▼──────┐ ┌─▼──────┐ ┌─▼──────┐
      │  Supabase    │ │ Paystack │ │ Resend  │
      │  (Postgres)  │ │ (Payments)│ │ (Email) │
      │  + Auth      │ │         │ │        │
      └──────┬───────┘ └────────┘ └────────┘
             │
      ┌──────▼──────┐
      │ Gemini AI   │
      │ (Lesson     │
      │  content)   │
      └────────────┘
```

Route Groups

GROUP	PATH	AUTH	PURPOSE
`(auth)`	/login, /register, /verify, /forgot-password	Public	Authentication
`(dashboard)`	/dashboard, /courses, /progress, /gpa, /capstone	Required + onboarding	Student university app
`(classroom)`	/classroom/[courseId]	Required	Full-screen course viewer
`(admin)`	/admin/**	Required (admin guard)	Admin CMS
`(courses-app)`	/courses-app/**	Optional	Standalone courses platform
`university`	/university/**	Public/Required	University-specific pages
root	/, /pricing, /onboarding, /about	Public	Landing, pricing, onboarding

Core Features Summary

FEATURE	STATUS	NOTES
Auth (email/password + verification)	Complete	Supabase Auth with email redirect
Student onboarding	Complete	8 departments
4-year curriculum with progression	Complete	Years 100→400, semester unlocks
Quiz/exam engine with grading	Complete	Edge Function for server-side grading
GPA calculation (5.0 scale)	Complete	Semester + cumulative GPA
Capstone submission + review	Complete	Premium + 400-level + GPA ≥ 3.5
Community suggestions system	Complete	Vote, review, auto-promote contributors
Premium subscriptions (Paystack)	Complete	Semester/monthly/yearly/lifetime plans
Standalone courses (pay-per-course)	Complete	YouTube/Drive lessons, lifetime access
Admin CMS	Complete	Multi-level admin permissions
Audit logging	Complete	Immutable audit trail
AI lesson content (Gemini)	Complete	Summaries, related content, deep-dives
Email (Resend)	Complete	Welcome, password reset, purchase confirmation
Demo mode	Complete	Cookie-based toggle, hidden in production
Account suspension	Complete	Admin can suspend/unsuspend
Transcript PDF generation	Complete	Server-side PDF streaming
Partnership inquiries	Complete	Form + database storage

Database Tables (25+)

CATEGORY	TABLES
Core	profiles, years, semesters, courses, lessons, resources, questions
Assessment	quiz_attempts, course_grades, semester_gpas, cumulative_gpas
Progression	enrollments, user_progress, semester_progress, course_prerequisites
Capstone	capstone_submissions, specialization_tracks, course_tracks
Community	suggestions, suggestion_votes, lesson_reactions, lesson_comments
Monetization	plans, subscriptions, transactions, standalone_courses, standalone_purchases
Admin	audit_log, notifications, webhook_events, partnership_inquiries
AI Content	lesson_summaries, lesson_deep_dive_links

Key Metrics

- **8 departments** in the university program
- **31+ courses** seeded in the curriculum
- **4 academic years** (100–400 level)
- **2 semesters** per year
- **5.0 GPA scale** with 8 grade points (A+ through F)
- **4 admin levels** (super, content, finance + student role)
- **3 plan types** (semester, monthly, yearly + lifetime)
- **2 video sources** (YouTube, Google Drive)

2. User Stories

Authentication & Onboarding

AS-001 — As a visitor, I want to browse the landing page and pricing so that I can understand what Aorthar Academy offers before signing up.

AS-002 — As a visitor, I want to create an account with my email, name, and password so that I can access the platform.

AS-003 — As a new student, I want to verify my email so that I can activate my account.

AS-004 — As a university student, I want to select my department during onboarding so that I see the correct curriculum for my field.

AS-005 — As a returning user, I want to log in with my email and password so that I can access my account.

AS-006 — As a user who forgot my password, I want to reset it via email so that I can regain access to my account.

University Student App

US-001 — As a university student, I want to see my dashboard with my current semester, courses, and progress so that I know what to study next.

US-002 — As a student, I want to view my course list so that I can see all available courses for my department and level.

US-003 — As a student, I want to enter the classroom for a course so that I can study lessons and take quizzes.

US-004 — As a student, I want to take a quiz so that I can test my knowledge and earn a grade.

US-005 — As a student, I want to take an exam so that I can complete the course assessment.

US-006 — As a student, I want to see my GPA so that I can track my academic performance.

US-007 — As a student, I want to see my progress across semesters so that I know what I've completed and what's next.

US-008 — As a 400-level student with premium access and GPA ≥ 3.5, I want to submit my capstone project so that I can graduate.

US-009 — As a student, I want to suggest improvements to the curriculum so that I can contribute to the community.

US-010 — As a student, I want to upgrade to premium so that I can access Year 400 courses and the capstone.

Standalone Courses

SC-001 — As a visitor, I want to browse standalone courses so that I can find courses to purchase.

SC-002 — As a visitor, I want to preview the first lesson of a course for free so that I can evaluate the course quality.

SC-003 — As a logged-in user, I want to purchase a course so that I get lifetime access.

SC-004 — As a course buyer, I want to watch purchased lessons so that I can learn at my own pace.

SC-005 — As a learner, I want to see my purchased courses in one place so that I can easily resume learning.

Admin Dashboard

AD-001 — As an admin, I want to see an overview dashboard so that I can quickly understand platform health.

AD-002 — As an admin, I want to manage courses so that I can create, edit, and publish curriculum content.

AD-003 — As an admin, I want to manage lessons and resources within a course so that I can structure the learning content.

AD-004 — As an admin, I want to manage the question bank so that I can create quiz and exam questions.

AD-005 — As an admin, I want to view and manage users so that I can see who has registered, their roles, and their purchases.

AD-006 — As an admin, I want to view all transactions so that I can monitor revenue and payment history.

AD-007 — As an admin, I want to review student suggestions so that I can approve or reject curriculum improvements.

AD-008 — As an admin, I want to review capstone submissions so that I can approve or request revisions.

AD-009 — As an admin, I want to manage standalone courses so that I can create and edit pay-per-course offerings.

AD-010 — As a super admin, I want to manage pricing plans so that I can adjust subscription pricing.

AD-011 — As a super admin, I want to view audit logs so that I can track all admin actions on the platform.

AD-012 — As a super admin, I want to suspend a user's account so that I can block access for policy violations.

AD-013 — As a super admin, I want to grant premium access to a user so that I can give them free premium status.

AD-014 — As a super admin, I want to grant standalone course access to a user so that I can give them a course without payment.

Settings & Account

ST-001 — As a user, I want to update my profile information so that my name and social links are current.

ST-002 — As a student, I want to change my department after onboarding so that I can switch to a different curriculum.

ST-003 — As a user, I want to delete my account so that I can remove all my data from the platform.

ST-004 — As a user, I want to download my academic transcript so that I have a record of my performance.

Notifications & System

NT-001 — As a student, I want to receive a welcome email after signing up so that I feel welcomed and know next steps.

NT-002 — As a course buyer, I want to receive a purchase confirmation email so that I have a record of my purchase.

NT-003 — As a student, I want to receive a password reset email so that I can recover my account.

Partnership

PT-001 — As a potential partner, I want to submit a partnership inquiry so that I can explore collaboration opportunities.

3. Module PRDs

Module 1: Authentication & Onboarding

Overview

Email/password authentication with email verification, password reset, and department-based onboarding for university students. Standalone course users skip department selection.

Goals

- Frictionless signup with email verification
- Automatic profile creation via database trigger
- Department routing for university students
- Cross-subdomain session sharing (.aorthar.com cookie domain)

User Flows

```
University Signup:
Visit aorthar.com/register → Fill name, email, password, department →
Verify email → Onboarding → Dashboard

Standalone Signup:
Visit courses.aorthar.com → Click "Sign up & enroll" → Fill name, email, password →
Verify email → Redirect to checkout → Paystack → Course access

Login:
Visit /login → Enter email + password → Redirect to ?next or default dashboard
```

Functional Requirements

ID	REQUIREMENT
FR-001	Sign up with email, password (8+ chars, uppercase, number), full name
FR-002	Email verification required before access
FR-003	Password reset via email link
FR-004	Auto-create profile row via handle_new_user() trigger
FR-005	Department selection on university domain (8 options)
FR-006	Minimal signup on courses domain (no department)
FR-007	Session persistence across subdomains via .aorthar.com cookie
FR-008	Suspended users redirected to /suspended

API / Data Requirements

ENDPOINT	STATUS	PURPOSE
`POST /api/auth/callback`	Real	Supabase email verification
`POST /api/auth/forgot-password`	Real	Send reset email via Resend
`POST /api/auth/send-welcome`	Real	Send welcome email via Resend
Supabase Auth	Real	signInWithPassword, signUp, updateUser

Module 2: University Student App

Overview

The core university experience: 4-year curriculum, quiz/exam engine, GPA tracking, progression gates, and capstone submission.

Goals

- Structured learning path through department-specific curriculum
- Fair assessment with server-side grading and cooldowns
- Accurate GPA tracking on 5.0 scale
- Clear progression gates (semester/year unlocks)

Functional Requirements

ID	REQUIREMENT
FR-001	Display courses by semester with status indicators
FR-002	Classroom with lessons, materials, related content, quiz tabs
FR-003	Timed quiz/exam with shuffled questions, no answer leakage
FR-004	Server-side grading via Supabase Edge Function
FR-005	Cooldown between attempts (default 24 hours)
FR-006	Attempt limits (default 3 per quiz/exam)
FR-007	Grade computation: quiz_weight × quiz_score + exam_weight × exam_score
FR-008	GPA on 5.0 scale with 8 grade points
FR-009	Semester unlock: all previous semester courses must pass
FR-010	Year 400 requires premium subscription
FR-011	Capstone requires 400-level + GPA ≥ 3.5 + premium

Module 3: Standalone Courses

Overview

Pay-per-course platform on courses.aorthar.com. One-time purchase via Paystack, lifetime access, YouTube/Drive-powered lessons.

Goals

- Simple purchase flow with minimal friction
- Free preview to drive conversions
- Reliable purchase recording (webhook + redirect fallback)
- Progress tracking across lessons

User Flows

```
Purchase Flow:
Browse courses → Select course → Watch 1-min preview →
Click "Buy" → Paystack checkout → Payment success →
Redirect to /courses-app/learn/[slug] → Record purchase →
Access all lessons
```

Functional Requirements

ID	REQUIREMENT
FR-001	Display published standalone courses with pricing
FR-002	Free 1-minute preview of first lesson
FR-003	Paystack checkout with course metadata
FR-004	Idempotent purchase recording (upsert on paystack_reference)
FR-005	Purchase verification on redirect AND via webhook
FR-006	YouTube and Google Drive video embedding
FR-007	Lesson progress tracking
FR-008	"My Courses" page showing all purchased courses

Module 4: Admin Dashboard

Overview

Multi-level admin CMS for managing courses, users, payments, pricing, curriculum, and platform operations.

Goals

- Complete content management for university and standalone courses
- User management with suspension and premium controls
- Financial visibility (transactions, revenue, conversion rates)
- Audit trail for all admin actions
- Role-based access control (super, content, finance)

User Flows

```
Course Creation:
Admin → /admin/courses → Create course → Set code, name, year, semester →
Add lessons → Add resources → Add questions → Publish course

User Management:
Admin → /admin/users → Search/filter users → View user details →
Suspend/unsuspend → Grant premium → Grant standalone access

Payment Monitoring:
Admin → /admin/payments → View unified transaction table →
See subscriptions + standalone purchases → Filter by type →
Monitor revenue and conversion rates
```

Functional Requirements

ID	REQUIREMENT
FR-001	CRUD for courses, lessons, resources, questions
FR-002	User listing with search, role filter, premium/purchase status
FR-003	Transaction listing (subscriptions + standalone purchases)
FR-004	Suggestion review with approve/reject
FR-005	Capstone review with approve/reject/request revision
FR-006	Standalone course management (create, edit, publish, add lessons)
FR-007	Pricing plan configuration
FR-008	Audit log viewer (immutable)
FR-009	User suspension/unsuspension
FR-010	Premium grant/revoke
FR-011	Standalone course access grant
FR-012	Student invite and bulk import
FR-013	Admin access management (invite, grant, update level)

Module 5: Payments & Monetization

Overview

Paystack-powered payment processing for university subscriptions and standalone course purchases. Webhook-based verification with idempotent recording.

Goals

- Reliable payment processing with webhook fallback
- Idempotent transaction recording
- Clear financial reporting for admins
- Multiple plan types (semester, monthly, yearly, lifetime)

User Flows

```
University Subscription:
Student → /pricing → Select plan → /pricing/checkout/[planId] →
Paystack checkout → Payment success → Webhook fires →
Edge Function verifies → Subscription created → Access granted

Standalone Course:
Student → /courses-app/[slug] → Click "Buy" →
/api/standalone/checkout → Paystack → Payment success →
Redirect to /courses-app/learn/[slug] → Verify payment →
Record purchase → Access all lessons
```

Functional Requirements

ID	REQUIREMENT
FR-001	Paystack checkout session creation
FR-002	Webhook signature verification (HMAC-SHA512)
FR-003	Idempotent upsert on paystack_reference
FR-004	Subscription status tracking (active/expired/cancelled)
FR-005	Purchase confirmation emails
FR-006	Revenue calculation and display
FR-007	Conversion rate calculation

Module 6: Settings & Account

Overview

User profile management, department changes, account deletion, and transcript downloads.

Functional Requirements

ID	REQUIREMENT
FR-001	Edit full name, bio, GitHub URL, LinkedIn URL
FR-002	Change department (recalculates progress)
FR-003	Delete account (cascading delete)
FR-004	Download transcript PDF

Module 7: Notifications & Email

Status: Partial

Email delivery via Resend for welcome, password reset, and purchase confirmation. In-app notifications table exists but is not yet surfaced in the UI.

- Email templates exist and work
- `notifications` table exists in schema but has no UI
- No push notifications or SMS

Module 8: University Pages

Status: Partial

Separate university-facing pages at `/university/*` for open-source university-like courses. Distinct from the standalone courses platform.

- Pages exist but integration with main dashboard is unclear
- Separate from `(dashboard)` route group
- May be legacy or parallel product arm

4. QA Test Cases

Smoke Test Suite (Critical Path)

#	TEST	MODULE	PRIORITY
SM-01	User can register with valid credentials	Auth	High
SM-02	User receives verification email	Auth	High
SM-03	User can verify email and access dashboard	Auth	High
SM-04	User can log in with valid credentials	Auth	High
SM-05	Dashboard loads with courses	University	High
SM-06	User can start and submit a quiz	University	High
SM-07	User can purchase a standalone course	Standalone	High
SM-08	Admin can view users and transactions	Admin	High
SM-09	Payment webhook records purchase	Payments	High
SM-10	Suspended user cannot access platform	Auth	High

Detailed Test Cases — Authentication

TC	TITLE	PRIORITY	EXPECTED
AUTH-01	Register with valid university credentials	High	Account created, redirected to /verify, email sent
AUTH-02	Register with valid standalone credentials	High	Account created, redirected to /verify, email sent
AUTH-03	Register with weak password	Medium	Error: "Must contain at least one uppercase letter"
AUTH-04	Register with existing email	Medium	Supabase error shown to user
AUTH-05	Register with invalid email	Low	Error: "Invalid email address"
AUTH-06	Login with correct credentials	High	Redirected to dashboard
AUTH-07	Login with wrong password	Medium	Error shown, no redirect
AUTH-08	Login with unverified email	Medium	Login succeeds but onboarding gate blocks access
AUTH-09	Forgot password flow	High	Reset email sent, success message shown
AUTH-10	Reset password	High	Password updated, redirected to dashboard
AUTH-11	Reset password with mismatched passwords	Low	Error: "Passwords do not match"
AUTH-12	Suspended user login	High	Redirected to /suspended
AUTH-13	Session persistence across subdomains	High	User is authenticated on both domains
AUTH-14	Logout from standalone platform	Medium	Session cleared, redirected

Detailed Test Cases — University Student App

TC	TITLE	PRIORITY	EXPECTED
UNI-01	Dashboard loads with semester courses	High	Shows current year/semester, courses, progress
UNI-02	Course listing shows all semesters	Medium	Courses grouped by semester with status
UNI-03	Enter classroom for a course	High	Classroom loads with lessons, materials tabs
UNI-04	Start quiz	High	Quiz starts, timer begins, questions shown
UNI-05	Submit quiz	High	Score calculated, results displayed
UNI-06	Quiz answers not exposed to client	High	No correct answers in API response
UNI-07	Quiz cooldown enforcement	Medium	Blocked with cooldown timer
UNI-08	Quiz attempt limit enforcement	Medium	Blocked, no new attempt created
UNI-09	GPA page shows correct calculation	High	GPA matches manual calculation
UNI-10	Progress page shows semester status	Medium	Visual breakdown of completed/locked courses
UNI-11	Capstone submission (eligible student)	High	Status changes to "pending"
UNI-12	Capstone submission (ineligible student)	Medium	Form disabled or blocked
UNI-13	Submit curriculum suggestion	Medium	Suggestion recorded as pending
UNI-14	Premium subscription purchase	High	Subscription activated, premium features unlocked
UNI-15	Semester unlock after passing all courses	High	Semester 2 courses unlocked

TC	TITLE	PRIORITY	EXPECTED
UNI-16	Year 400 locked without premium	Medium	Blocked, prompted to upgrade
UNI-17	Department change	Medium	Profile updated, progress recalculated
UNI-18	Download transcript	Medium	PDF downloaded with grades and GPA
UNI-19	Delete account	High	Account deleted, logged out, data removed

Detailed Test Cases — Standalone Courses

TC	TITLE	PRIORITY	EXPECTED
SC-01	Browse published courses	High	Published courses displayed with prices
SC-02	View course detail page	High	Course page with preview, description, buy button
SC-03	Free preview (1 minute)	High	Preview ends, paywall overlay shown
SC-04	Purchase course via Paystack	High	Redirected to learn page, purchase recorded
SC-05	Purchase recording (webhook)	High	Purchase recorded in database
SC-06	Access purchased lessons	High	All lessons accessible, no paywall
SC-07	Access unpurchased course	Medium	Redirected to course page
SC-08	Lesson progress tracking	Medium	Previous lesson marked complete
SC-09	My Courses page	Medium	List of all purchased courses with progress
SC-10	Duplicate purchase prevention	Medium	Redirected to learn page (409)
SC-11	YouTube video playback	Medium	Video plays correctly
SC-12	Google Drive video playback	Medium	Video plays correctly
SC-13	Course not found (invalid slug)	Low	404 page
SC-14	Draft course not visible	Low	Course not shown

Detailed Test Cases — Admin Dashboard

TC	TITLE	PRIORITY	EXPECTED
AD-01	Admin dashboard loads	High	Metrics displayed: users, subscriptions, revenue
AD-02	Create course	High	Course created, appears in list
AD-03	Edit course	Medium	Changes saved and reflected
AD-04	Delete course	Medium	Course removed (or blocked if has enrollments)
AD-05	Add lesson to course	Medium	Lesson appears in course
AD-06	Add question to course	Medium	Question saved with correct options format
AD-07	View users list	High	All users shown with name, email, role, status
AD-08	Search users	Medium	Filtered results
AD-09	Filter users by role	Medium	Only matching users shown
AD-10	View standalone course buyers	High	Buyers shown with course names in Purchases column
AD-11	View payments/transactions	High	All subscriptions + standalone purchases shown
AD-12	Revenue calculation	Medium	Matches sum of all successful transactions
AD-13	Review suggestion	Medium	Status updated, contributor promoted if applicable
AD-14	Review capstone	Medium	Status updated, student notified
AD-15	Manage standalone courses	High	Course created and published

TC	TITLE	PRIORITY	EXPECTED
AD-16	Suspend user	High	User redirected to /suspended on next login
AD-17	Grant premium access	Medium	User gains premium features immediately
AD-18	Grant standalone access	Medium	User can access course without payment
AD-19	View audit logs	Medium	Chronological log of admin actions
AD-20	Non-admin access blocked	High	Redirected to /unauthorized (in staging/prod)
AD-21	Finance admin cannot edit courses	Medium	Blocked by permissions
AD-22	Content admin cannot view payments	Medium	Blocked by permissions

Detailed Test Cases — Payments

TC	TITLE	PRIORITY	EXPECTED
PAY-01	Paystack checkout session created	High	Payment URL returned
PAY-02	Webhook signature verification	High	400 response for invalid signature
PAY-03	Webhook records subscription	High	Subscription created in DB
PAY-04	Webhook records standalone purchase	High	Purchase recorded in DB
PAY-05	Idempotent webhook handling	High	Only one record created
PAY-06	Failed payment handling	Medium	No subscription/purchase created
PAY-07	Purchase confirmation email sent	Medium	Email received with course details
PAY-08	Subscription expiration	Medium	Status changed to expired
PAY-09	Premium access revoked on expiry	High	Blocked

Detailed Test Cases — Security

TC	TITLE	PRIORITY	EXPECTED
SEC-01	RLS prevents cross-user data access	High	No data returned
SEC-02	Admin API requires admin role	High	401/403 response
SEC-03	Quiz answers not leaked	High	No is_correct in options
SEC-04	Suspended user API blocked	High	403 response
SEC-05	Paystack webhook HMAC verification	High	400 response
SEC-06	Service role key not exposed	High	No SUPABASE_SERVICE_ROLE_KEY in bundle
SEC-07	Demo mode hidden in production	Medium	No demo toggle visible

Regression Test Scenarios

#	SCENARIO	PRIORITY
REG-01	After deploying new course, existing students see it	High
REG-02	After changing pricing, existing subscriptions unaffected	High
REG-03	After department change, student's progress recalculated	High
REG-04	After admin suspension, user immediately blocked	High
REG-05	After webhook reprocessing, no duplicate transactions	High
REG-06	After updating quiz questions, existing attempts unaffected	Medium
REG-07	After adding new lesson, course progress not reset	Medium

5. Gap Analysis

Critical Gaps

ID	GAP	IMPACT	MODULE	RECOMMENDATION
G-01	`notifications` table exists but has no UI	Medium	Notifications	Build notification bell in Navbar + notification list page
G-02	No email notification when suggestion is approved/rejected	Medium	Suggestions	Send email to proposer on status change
G-03	No email notification when capstone is reviewed	Medium	Capstone	Send email to student on review
G-04	`?module=courses` query param on admin pages has no effect	Low	Admin	Either implement filtering or remove the param from URLs
G-05	No pagination on admin users/transactions tables (limit 100)	Medium	Admin	Add pagination or infinite scroll for large datasets
G-06	No export functionality for transactions/users	Medium	Admin	Add CSV export for financial reporting
G-07	`univeristy/transaction` typo route exists	Low	Routing	Redirect to correct `/university/transactions` path
G-08	No real-time quiz submission confirmation	Low	University	Add loading state and explicit success/error feedback

Important Gaps

ID	GAP	IMPACT	MODULE
G-09	No bulk email capability for admins	Medium	Admin
G-10	No analytics dashboard (page views, engagement, drop-off)	Medium	Admin
G-11	No referral/affiliate system for standalone courses	Low	Standalone
G-12	No certificate of completion for standalone courses	Medium	Standalone
G-13	No search functionality for courses	Medium	Both
G-14	No mobile app or PWA support	Low	Platform
G-15	No discussion forum or community feature	Medium	University
G-16	No progress email reminders for inactive students	Low	Engagement
G-17	No A/B testing framework for pricing page	Low	Monetization
G-18	Standalone course purchase recording has 3 code paths	Medium	Standalone

Backend Gaps

ID	GAP	IMPACT	RECOMMENDATION
BG-01	No rate limiting on API routes	Medium	Add rate limiting for auth, quiz, and payment endpoints
BG-02	No API versioning	Low	Add `/api/v1/` prefix for future compatibility
BG-03	No webhook retry logic	Medium	If webhook processing fails, no automatic retry
BG-04	No background job queue	Low	Email sending is fire-and-forget; consider a queue
BG-05	No database backups automation	High	Ensure Supabase automated backups are enabled
BG-06	No migration rollback capability	Medium	Supabase migrations are forward-only

6. Technical Debt Analysis

Structural Issues

ID	ISSUE	SEVERITY	FIX
TD-01	**Triple purchase recording paths** — standalone_purchases recorded via 3 code paths creating maintenance burden	High	Consolidate to single source of truth
TD-02	**Duplicate university pages** — /university/* routes exist alongside (dashboard) routes	Medium	Clarify purpose; remove or integrate
TD-03	**Typo in route** — univeristy/transaction (missing 's')	Low	Add redirect to correct path, then remove
TD-04	**Inconsistent timestamp columns** — purchased_at vs created_at vs paid_at	Medium	Standardize naming convention
TD-05	**AI library misnamed** — openai.ts actually calls Gemini API	Low	Rename to gemini.ts or ai.ts

Security Risks

ID	RISK	SEVERITY	MITIGATION
TD-21	**Admin guard disabled in development** — NEXT_PUBLIC_APP_ENV=development bypasses admin check	High	Use server-side env check, not client-side variable
TD-22	**No rate limiting** — Auth, quiz, and API endpoints have no rate limiting	Medium	Add rate limiting (Upstash Ratelimit or Vercel Edge Middleware)
TD-23	**Paystack metadata not validated server-side**	Medium	Add server-side validation of metadata
TD-24	**No CSRF protection on forms**	Low	Supabase handles this via session cookies, but verify
TD-25	**Audit log IP address may be inaccurate behind proxy**	Low	Use x-forwarded-for header parsing

Scalability Risks

ID	RISK	SEVERITY	MITIGATION
TD-16	**No pagination on admin tables** — limit 100 will break at scale	Medium	Add cursor-based pagination
TD-17	**No database indexing on hot query paths**	Medium	Verify indexes; add composite indexes
TD-18	**Fire-and-forget email sending** — emails lost if Resend is down	Medium	Add retry logic or use a queue
TD-19	**No CDN caching for static assets**	Low	Configure Vercel Image Optimization
TD-20	**Single Edge Function for grading**	Low	Monitor execution time; consider batching

Code Quality

ID	ISSUE	SEVERITY	FIX
TD-26	**No test coverage** — Vitest configured but no tests exist	High	Add unit tests for core logic
TD-27	**Type assertions (as unknown)** — Fragile Supabase join casting	Medium	Create proper TypeScript types
TD-28	**Large page components** — Some files exceed 200 lines	Low	Extract data fetching into separate functions
TD-29	**No error boundaries**	Medium	Add error boundaries at page and layout level
TD-30	**Console.error used for production errors** — No monitoring	Medium	Integrate Sentry or similar error tracking

Priority Fix Order

1. TD-21 — Admin guard bypass risk (security)
2. TD-26 — No test coverage (quality)
3. TD-01 — Triple purchase recording (reliability)
4. TD-22 — No rate limiting (security)
5. TD-16 — No pagination on admin tables (scalability)
6. TD-18 — Fire-and-forget emails (reliability)
7. TD-04 — Inconsistent timestamp columns (maintainability)
8. TD-30 — No error monitoring (observability)

7. One-Year Roadmap

Q1 (Apr–Jun 2026) — Stabilization & Core Fixes

Goals

Fix all critical bugs and gaps. Establish testing foundation. Improve admin experience for monitoring buyers and transactions.

FEATURE	PRIORITY
Admin transaction visibility — ensure all standalone buyers appear in users list and payments page	High
Admin pagination — Add pagination to users, transactions, and suggestions tables	High
CSV export — Export users and transactions from admin panel	High
Email notifications for reviews — Notify students when suggestions/capstones are reviewed	High
Rate limiting — Add rate limiting to auth, quiz, and payment endpoints	High
Error monitoring — Integrate Sentry for production error tracking	High
Test suite — Add unit tests for GPA calculation, progression logic, and grading	High
Fix typo route — Redirect /univeristy/transaction to correct path	Medium

Success Metrics: 0 critical bugs in production · Test coverage > 40% · Admin page load time < 2s with 1000+ users · Email delivery rate > 99%

Q2 (Jul–Sep 2026) — Core Feature Expansion

Goals

Enhance the learning experience for both university and standalone students. Build community features. Improve course discovery.

FEATURE	PRIORITY
Course search — Full-text search across university and standalone courses	High
Discussion forums — Course-level discussion threads for university courses	High
Certificate of completion — PDF certificates for standalone courses	High
Notification system — Build UI for existing notifications table	High
Course reviews/ratings — Star ratings and written reviews for standalone courses	Medium
Progress email reminders — Weekly nudges for inactive students	Medium
Bulk email for admins — Send announcements to all students or segments	Medium
Coupon/discount codes — Promo code system for marketing campaigns	Medium

Success Metrics: 20% increase in course completion rate · 15% increase in standalone course conversions · 50+ discussion threads active per month

Q3 (Oct–Dec 2026) — Growth & Monetization

Goals

Drive revenue growth through new monetization features. Expand standalone course catalog. Build referral and affiliate systems.

FEATURE	PRIORITY
Referral program — Referral codes for standalone courses with discounts	High
Affiliate dashboard — Track affiliate earnings and referrals	High
Instructor onboarding — Allow external instructors to create and manage standalone courses	High
Course bundles — Bundle multiple standalone courses at a discount	Medium
Gift courses — Purchase courses as gifts for other users	Medium
Multi-currency support — Support USD in addition to NGN	Medium
A/B testing framework — Test pricing page layouts and copy	Medium
Analytics dashboard — Page views, engagement metrics, drop-off analysis	High

Success Metrics: 30% of standalone revenue from referrals/affiliates · 5+ external instructors onboarded · 20% increase in ARPU

Q4 (Jan–Mar 2027) — Scale & Optimization

Goals

Prepare for 10x user growth. Optimize performance and reliability. Explore new product opportunities.

FEATURE	PRIORITY
PWA support — Offline lesson viewing for standalone courses	High
Mobile app — React Native app for iOS and Android	Medium
Gamification — Streaks, badges, leaderboards for university students	Medium
Social sharing — Share certificates and achievements on LinkedIn/Twitter	Low
Advanced analytics — Cohort analysis, LTV prediction, churn modeling	Medium
API for partners — Public API for university partnerships	Medium
White-label option — Allow institutions to white-label the platform	Low
AI tutor — Gemini-powered Q&A assistant for course content	Medium

Success Metrics: 10,000+ active students · 99.9% uptime · < 1s page load time globally · 50+ courses in standalone catalog

8. Product Recommendations

High-Impact, Low-Effort (Do First)

#	RECOMMENDATION	IMPACT	EFFORT
1	**Admin CSV Export** — Add export button to users and payments pages	High	2–4 hours
2	**Email Notifications for Reviews** — Notify students on suggestion/capstone review	High	4–6 hours
3	**Admin Pagination** — Add pagination to all admin tables	High	4–8 hours
4	**Consolidate Purchase Recording** — Single source of truth for standalone purchases	High	8–12 hours
5	**Rate Limiting** — Protect auth, quiz, and payment endpoints	High	4–8 hours

High-Impact, Medium-Effort (Do Next)

#	RECOMMENDATION	IMPACT	EFFORT
6	**Course Search** — Full-text search with filters	High	1–2 weeks
7	**Certificate of Completion** — PDF certificates for standalone courses	High	1 week
8	**Notification System UI** — Bell icon + notification list	Medium	1–2 weeks
9	**Coupon/Discount Codes** — Promo codes for marketing	High	1–2 weeks
10	**Analytics Dashboard** — PostHog integration	High	1 week

Strategic Recommendations

#	RECOMMENDATION	WHY
16	**Instructor Marketplace** — Allow external instructors to sell courses	Transforms from single-instructor to marketplace, dramatically increasing catalog and revenue
17	**Corporate/B2B Licensing** — Bulk licenses for company teams	Higher ARPU than individual sales
18	**Mobile App (PWA First)** — Offline lesson viewing	Lower development cost, instant distribution, works on all devices
19	**AI Tutor** — Gemini-powered Q&A assistant	Differentiates from competitors, increases learning outcomes
20	**White-Label for Universities** — B2B SaaS model	Recurring revenue from institutional partnerships

Quick Wins (Under 2 Hours Each)

#	CHANGE	IMPACT
QW-01	Fix /univeristy/transaction typo redirect	Eliminates 404s
QW-02	Rename src/lib/ai/openai.ts to gemini.ts	Reduces confusion
QW-03	Add rel="noopener noreferrer" to all external links	Security best practice
QW-04	Add Open Graph meta tags to all public pages	Better social sharing
QW-05	Add loading skeletons to admin tables	Perceived performance
QW-06	Add empty state illustrations	Better UX
QW-07	Standardize date formatting across all pages	Consistency
QW-08	Add ?module=courses filtering logic or remove param	Eliminates confusion

9. Scope Control

In Scope (Current Product)

- Email/password authentication with verification
- Password reset via email
- Student onboarding (department selection)
- 4-year university curriculum (100–400 level)
- 8 departments with semester-based progression
- Quiz/exam engine with server-side grading
- GPA calculation (5.0 scale)
- Capstone submission and review
- Community suggestions system
- Premium subscriptions (Paystack)
- Standalone courses (pay-per-course)
- YouTube and Google Drive video embedding
- Admin CMS (courses, users, payments, pricing)
- Multi-level admin permissions (super, content, finance)
- Audit logging
- Account suspension
- Email delivery (Resend)
- AI lesson content (Gemini)
- Demo mode
- Transcript PDF generation
- Partnership inquiries

Out of Scope (Explicitly Not Building)

CATEGORY	ITEMS
Infrastructure	Live video streaming, video hosting/encoding, CDN infrastructure, database infrastructure
Product Features	Real-time collaboration, blockchain certificates, multi-language support, native mobile apps, corporate/B2B sales, white-label platform, discussion forums, referral/affiliate system, coupon codes, course reviews, gamification, AI tutor, instructor marketplace
Technical	API versioning, microservices architecture, database sharding, multi-region deployment

Grey Area (Needs Clarification)

ITEM	STATUS	NOTES
`/university/*` pages	Unclear	Separate product arm or legacy? Needs product decision
`?module=courses` query param	Unused	Present in URLs but not implemented. Implement or remove.
In-app notifications UI	Table exists, no UI	Build or remove?
External instructor accounts	Not yet designed	Instructor name is hardcoded. Should instructors have their own accounts?
Course sort order	Partial	sort_order column added but not fully utilized in UI